Dehazing Attention GAN: Channel-Spatial U-Net Architecture for Robust Image Dehazing and Object Detection

D. Pushpalatha*
Department of Electronics and Communication Engineering
National Institute of Technology,
Warangal, 506004, Telangana, India
dp23ecr2r16@student.nitw.ac.in,
*- Corresponding Author.

P. Prithvi
Department of Electronics and Communication Engineering,
National Institute of Technology,
Warangal, 506004, Telangana, India
prithvi@nitw.ac.in.

***Abstract* - Image dehazing plays an essential role in enhancing the clarity, accuracy, and overall quality of visual data captured in hazy environments. At the same time, deep learning approaches have replaced traditional statistical methods; challenges remain in restoring color and fine details. This work introduces a Generative Adversarial Network (GAN)-based dehazing method using a U-Net-base generator enhanced with a Channel and Spatial Attention Residual (CSAR) block. The proposed design strengthens feature representation across both channel and spatial dimensions, reduces the computational cost, and helps prevent gradient vanishing during training. The effectiveness of the suggested Dehazing Attention GAN (DAGAN) in enhancing clarity and detection accuracy in hazy environments was demonstrated by its evaluation on multiple benchmark datasets and subsequent validation through object detection using YOLOv8. The results indicate that DAGAN generates images that are more suitable for autonomous vehicle applications in difficult visual environments by producing images with finer details and better structural consistency.**

Keywords— Channel and Spatial Attention Residual (CSAR), DAGAN (Dehazing Attention Generative Adversarial Network) Generative Adversarial Networks (GAN), Image restoration, Object detection.

I. Introduction

Haze, which is brought on by suspended particles scattering light, causes data loss and deterioration of camera images. This has implications for computer vision tasks such as segmentation [1]. Reducing the efficiency of image classification [3] and object detection [2]. Dehazing a single image is now essential. Several strategies have been developed using the ASM (Atmospheric Scattering Model) to increase the dependability of computer vision applications [4].

$$I(x) = \frac{J(x) - A(1 - t(x))}{J(x)} \tag{1}$$

The hazy image is represented by J(x), and the corresponding clear (haze-free) image is represented by I(x). While t(x) denotes the transmission map at a specific pixel position x, 'A' denotes the global atmospheric light. Here

$$t(x) = e^{\beta d(x)} \tag{2}$$

Here, β stands for the scattering coefficient and d(x) for the distance between the object and the camera.

To restore clear images from hazy ones, a number of ASM-based techniques have been proposed [5] [6], with an emphasis on estimating "A" and t(x) for image restoration. Physical priors might not always hold, though, making it challenging to determine these with accuracy in the absence of trustworthy priors. ASM-based methods have trouble with severely degraded remote sensing images because haze density rises with depth, which frequently results in color distortion, artifacts, and inadequate dehazing.

Haze affects tasks like object detection, recognition, and tracking by lowering image contrast and clarity. Traditional methods, such as Color Attenuation Prior [8] and Dark Channel Prior [7], rely on strict assumptions and struggle with complex or uneven haze. By using end-to-end learning, deep learning techniques enhance performance; however, they still have problems with redundant features, loss of fine details, and poor generalization under different conditions.

Key Contributions of This Work Include:

- A Generative Adversarial Network based image dehazing technique is proposed DAGAN Network, aimed at enhancing both color fidelity and fine detail restoration.
- Compact U-Net generator architecture is designed, incorporating a novel Channel and Spatial Attention Residual (CSAR) block to improve feature extraction and representation.
- Demonstrated notable improvements in object detection accuracy by integrating the dehazed outputs with the YOLOv8 framework.

II. RELATED WORK

This section examines other research conducted on GAN based single-image dehazing. Prior-based techniques, data-driven methods, and GAN frame works have all attempted to address the difficult task of recovering clear visuals from haze-affected images [9].

a) Prior – based Methods

The Atmospheric Scattering Model acts as the foundation for most dehazing methods (ASM) [10, 11], with many prior-based techniques developed under this framework [12–14].

2025 1st International Conference on Data Science and Intelligent Network Computing (ICDSINC) | 979-8-3315-6524-4/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICDSINC66221.2025.1448232

He et al. [20] introduced the Dark Channel Prior [21] observes that, in non-sky areas, one color channel is typically dark. While such priors enhance contrast and visibility, they often fail in diverse environments, resulting in color distortions and visual artifacts due to their limited generalizability.

b) Learning based methods

Deep learning-based dehazing has garnered significant attention because it can accurately model the intricate, nonlinear connections between hazy and clear images. Since Dehaze-Net [22], models like AOD-Net [23], FAMED-Net [13], and RCAN [14] have used data-driven approaches. Image dehazing techniques can be grouped into two main types: (i) approaches that utilize physical models to estimate factors such as atmospheric light and transmission maps, and (ii) methods that directly generate dehazed images without relying on explicit physical modeling.

c) GAN based methods

Goodfellow et al. first proposed GANs in 2014 [15], and they have since developed into an essential framework for image restoration tasks [16, 17]. The generator produces realistic outputs, while the discriminator distinguishes between the real and the fake. Adversarial training is used to refine both until they reach Nash equilibrium [21], at which point the results closely resemble actual information.

$$min_G max_D V(D,G) = E_{x\sim p_r}[\log(D(x))] + E_{\tilde{x}\sim p_g}[log(1 - D(\tilde{x}))] \quad (3)$$

In this context, G is the generator, D is the discriminator, and E denotes the expected value, x a real sample from P_r, and $\tilde{x} = G(z)$ a generated sample from p_g using random noise z. Training alternates between updating D to distinguish real from fake and G to fool D. D scores real data higher, while G learns to produce outputs that D can't easily reject.

$$min_G V(D,G) = E_{\tilde{x}\sim p_g}[log(1 - D(\tilde{x}))] \quad (4)$$

The generator's goal is to create data that closely mimic real samples, with higher D ($\tilde{x}$) values indicating better performance. Minimizing the generator's loss is crucial. As depicted in Fig. 1, training involves both hazy and clear images, enabling the generator to learn how to create realistic dehazed outputs. In order to help with convergence, the discriminator scores the generated images' realism.

III. PROPOSED METHOD

3.1 Generator Model Design Overview

The developed DAGAN framework is proposed as an effective solution for the image dehazing task utilizes the generative adversarial architecture comprising two primary neural networks. One responsible for generating images (the generator) and the other for evaluating them (the discriminator). The internal structure and design of the generator component are depicted in Fig. 2.

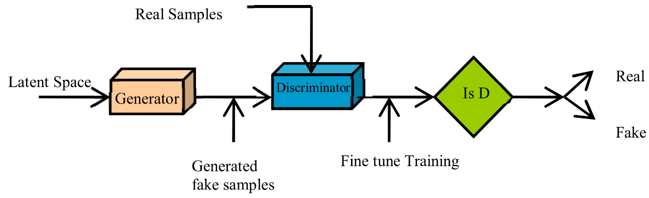

Fig.1. Illustrates the fundamental structure of Generative Adversarial Networks (GANs).

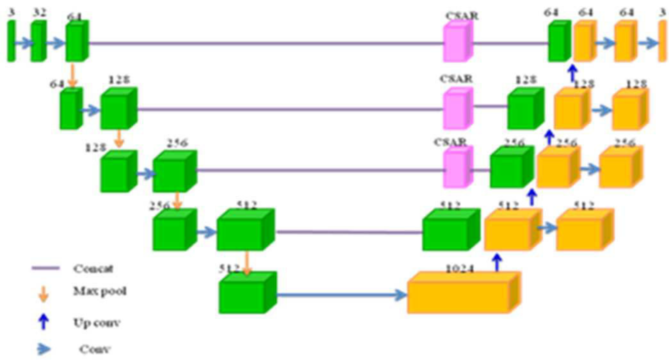

Fig.2. DAGAN Network: Generator Design Overview

The generator makes use of a U-Net-based architecture. For tasks like image dehazing, it is accurate and efficient because it captures both local details and global context.

The efficiency and simplicity of the U-Net architecture make it highly regarded. A reduced overlap approach was used to expedite processing, avoiding the standard sliding window methods' redundant computations. In this work, U-Net uses 3×3 convolutions and Leaky ReLU activations to improve learning while striking a balance between image context and accurate localization. Leaky ReLU permits the tiny gradients seen in Eq. (5), as opposed to ReLU, which can impede learning with zero gradients for negative inputs, guaranteeing more reliable training.

$$f(x) = max(0.01x, x) \quad (5)$$

Each 2×2 max pooling layer halves the spatial dimensions while doubling feature channels, increasing from 64 to 512 across the contracting path to capture abstract, contextual features. In the expanding path, 2×2 convolutions restore resolution and improve localization by up sampling and merging with earlier features. CSAR blocks are used instead of simple concatenation to better emphasize important spatial and channel-wise information.

3.2 Channel and Spatial Attention Residual Blocks

By concentrating on important spatial areas or channels, attention mechanisms-which are just as important as convolutions-improve image processing, much like human visual attention. To keep crucial details intact, they allocate adaptive weights according to feature relationships. Strong results have been obtained in previous work when spatial and channel attention are combined [25, 26].

Convolutions in image restoration convert RGB pictures into multi-channel features that aid in the reconstruction of minute details. The channel attention module maintains important channel information and reducing spatial dimensions to 1×1. By adding spatial attention after channel attention, the suggested CSAR block aids in the network's ability to concentrate on crucial areas, cut down on redundancy, and remember specifics throughout training. The network combines channel attention for non-local dependencies [24] and spatial attention for local features in order to prevent gradient vanishing and ensure stable learning. This enhances image restoration by better learning local details and global context

3.3 Discriminator Block

The discriminator distinguishes between real and fake photos. It evaluates the generator output, hazy input, and ground truth during dehazing. In order to improve perceptual realism and better capture fine details, this work uses a PatchGAN [25, 26] instead of analyzing the entire image. Every element of the NxN matrix produced by the PatchGAN discriminator indicates the probability that a particular local patch is real or fake.

Primary GAN discriminators are less appropriate for high-resolution tasks where meticulous details must be maintained because they only produce one scalar. Shared weights enhance performance, realism, and convergence speed; this regional focus is more effective at capturing fine textures than the traditional discriminators. In the proposed DAGAN Network, a PatchGAN with N=70 is used to average patch scores for the output. Group normalization [27] is used instead of batch normalization to ensure stable training with small batches and lower overhead. Five convolutional layers and Leaky ReLU are used in the discriminator to achieve smooth, steady convergence using Wasserstein loss.

IV. LOSS FUNCTION

A good loss function ensures effective training and convergence. This study uses a composite loss combining adversarial loss for textures and content loss for structure and semantics, enabling accurate haze-free reconstruction. The loss is defined as:

$$\mathrm{L_{total}} = \mathrm{L_{adv}} + \lambda \mathrm{L_p} \tag{6}$$

The adversarial loss promotes realism, while content loss preserves structure, balanced by λ = 100. Training is alternated between the generator and discriminator with hazy and clear images. Discriminator uses Wasserstein distance [18] for stable gradients.

$$L_D = E_{\tilde{x}\sim p_g}[D(\tilde{x})] - E_{x\sim p_r}[D(x)] + \lambda E_{\hat{x}\sim p_{\hat{x}}}[(\nabla_{\hat{x}} D(\hat{x})_2 - 1)]^2 \tag{7}$$

Here, $\tilde{X}$ ~Pg is the generator's output distribution, X ~Pr is the real clear image distribution, and $\tilde{X}$ ~P_X is the penalty distribution between output and ground truth. ∇ denotes the gradient operator, and λ controls penalty strength. As described in [18], the Wasserstein distance.

$$D(x) = \nabla_w \frac{1}{m}\sum_{i=1}^{m}[\mathrm{f}(x^{(i)} - f(G(z^{(i)}))] \tag{8}$$

In this formulation, *f(x)* is constrained to [–0.01, 0.01] to satisfy the 1-Lipschirz condition. The adversarial loss for the dehazing model is then defined as:

$$\mathrm{L_{adv}} = \sum_{n=1}^{N} - D_{\theta_D}\left(G_{\theta_G}(I^H)\right) \tag{9}$$

Here, *N* is the batch size, *D* the discriminator, *G* the generator, and I^H the hazy input. Content loss measures perceptual difference between output and clear reference, guiding the generator to preserve structure and context. It is formulated as:

$$L_P = \frac{1}{W_{i,j}H_{i,j}}\sum_{x=1}^{W_{i,j}}\sum_{y=1}^{H_{i,j}}(\phi_{i,j}(I^S)_{x,y} - \phi_{i,j}(G_{\theta_G}(I^H))_{x,y})^2 \tag{10}$$

Here, $W_{i,j}$ and $H_{i,j}$ are the feature map's and height, and $\phi_{i,j}$is the output from the j[th] convolution before the i[th] pooling layer of a pre-trained network. Adversarial and content losses together guided training. Over 200 epochs, the generator loss converged, showing reliable dehazing performance.

V. EXPERIMENTAL EVALUATION AND ANALYSIS

This section provides a thorough assessment of the DAGAN-based dehazing strategy, highlighting the leading methodologies and the superior efficacy and haze-elimination capabilities of the constructed model.

5.1 Performance Metrics

Dehazing performance is assessed subjectively as well as objectively using metrics. The industry-standard PSNR and SSIM metrics for image restoration assess both structural fidelity and image quality. PSNR uses Mean Squared Error (MSE) between the reference (I) and dehazed (K) images to guarantee a fair comparison of methods. Here's how to calculate the MSE:

$$MSE = \frac{1}{mn}\sum_{i=1}^{m-1}\sum_{j=1}^{n-1}(I(i,j) - K(i,j))^2 \tag{11}$$

Here, *i* and *j* denote pixel coordinates. From the computed MSE, PSNR is calculated as a quantitative measure of similarity between the dehazed and reference clear images:

$$PSNR = 10log_{10}\left(\frac{MAX_I^2}{MSE}\right) \tag{12}$$

Here, MAX^2_I is the square of the highest pixel intensity. For B-bit images, $MAX_I = 2^B - 1$. A higher PSNR denotes better restoration since it shows a closer resemblance between the clear and dehazed images.

Perceptual similarity is measured by SSIM, which evaluates luminance, contrast, and structure to demonstrate how well structural details are maintained.

$$SSIM(x,y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \quad (13$$

We display the average pixel intensities of the dehazed image x and the corresponding ground truth image y respectively, in this formulation. In order to capture the contrast information, the terms σ^2x and σ^2y stand for the variances of pixel values in the x and y directions. σ_{xy} signifies the covariance between the two images, which reflects the structural similarity. Constants $c_1$and c_2 are introduced to stabilize the division and prevent numerical instability when the denominators approach zero.

5.2 Training Details:

In this proposed DAGAN Network, we used a Tesla V100 GPU and PyTorch 1.12.0. Table 1 lists the specific hyper parameters used when training the DAGAN network.

Table.1. Configuration of Training Parameters for the DAGAN Network

S. No.	**Hyper-Parameter**	**Setting**
1	Input image Size	512 x 512
2	Optimizer	Adam
3	Epochs	200
4	Batch size	8
5	Learning rate	0.000001

5.3 Datasets

Several synthetic datasets were used to assess the DAGAN network's efficacy. The RESIDE dataset offers a wide selection of images that represents various levels of haze [28], was specifically used for this. In order to guarantee a variety of haze densities, environments, and lighting, we selected 5,000 samples at random for training and tested 1,000 samples from the ITS, OTS, SOTS, and HSTS subsets. In order to test the DAGAN model's resilience in practical settings; we employed a wide range of real hazy images that were collected from online databases. In addition, we trained and tested the DAGAN network using the following images: Indoor-Haze [29], Outdoor-Haze [30], Dense-Haze [31], and NH-Haze [32] (non-homogeneous haze).

5.4 Quantitative Comparison

The proposed DAGAN-based method was tested on both real and fake images against techniques such as DCP [33], DehazeNet [34], GMAN [35], GCANet [36], DCPDN [37], EPDN [38], HIDEGAN [39], GPGDN [40], and DPTE-Net [41]. All models were retrained with the same parameters to guarantee an equitable comparison. Quantitative results benchmark DAGAN network, with top scores in the Tables 2 and 3 has shown in bold to highlight superior performance.

Table.2. Quantitative comparison on RESIDE [28], I-HAZE [29], and O - HAZE [30] Datasets Based on PSNR and SSIM scores

Datasets	**RESIDE**		**I-Haze**		**O-Haze**	
Networks / Metrics	**PSNR (dB) ↑**	**SSIM ↑**	**PSNR (dB) ↑**	**SSIM ↑**	**PSNR (dB) ↑**	**SSIM ↑**
DCP[33]	21.569	0.875	17.67	0.825	19.21	0.841
DehazeNet[34]	21.127	0.902	19.23	0.837	20.25	0.867
GMAN[35]	22.327	0.892	19.21	0.843	30.35	0.916
GCANet[36]	21.729	0.875	27.32	0.897	23.56	0.901
DCPDN[37]	20.621	0.764	31.83	0.926	33.02	0.947
EPDN[38]	23.987	0.678	25.14	0.894	26.67	0.925
HIDEGAN[39]	24.713	0.868	25.02	0.872	27.89	0.932
GPGDN[40]	29.136	**0.972**	26.15	0.883	28.12	0.949
DPTE-Net[41]	29.465	0.911	28.23	0.915	28.41	0.942
Proposed(DAGAN)	**30.546**	0.961	**34.96**	**0.970**	**28.52**	**0.957**

Table.3. Quantitative comparison on Dense-Haze [31], and NH-Haze [32] Datasets Based on PSNR and SSIM scores

Datasets	**Dense-Haze**		**NH-Haze**	
Networks / Metrics	**PSNR (dB) ↑**	**SSIM ↑**	**PSNR (dB) ↑**	**SSIM ↑**
DCP[33]	13.25	0.674	14.23	0.712
DehazeNet[34]	15.23	0.704	16.12	0.657
GMAN[35]	18.09	0.767	19.45	0.743
GCANet[36]	18.16	0.778	18.31	0.733
DCPDN[37]	19.43	0.786	20.34	0.716
EPDN[38]	19.47	0.782	19.16	0.768
HIDEGAN[39]	20.06	0.782	22.24	0.774
GPGDN[40]	25.62	0.897	25.68	0.842
DPTE-Net [41]	25.83	**0.907**	24.12	0.871
Proposed(DAGAN)	**26.67**	0.903	**25.89**	**0.892**

5.5 Qualitative Comparison

On benchmark datasets such as RESIDE [28], Indoor-Haze [29], Outdoor-Haze [30], Dense-Haze [31], and NH-Haze [32], the suggested dehazing method continuously outperforms the leading techniques in the field demonstrate their effectiveness, which is illustrated in Figure 3 to 5, it effectively preserves image details and enhances clarity. With a PSNR of 34.96 dB and SSIM of 0.970 on randomly chosen samples, it obtained the highest ratings. Compared to nine leading methods DCP [33], DehazeNet [34], GMAN [35], GCANet [36], DCPDN [37], EPDN [38], HIDEGAN [39], GPGDN [40], and DPTE-Net [41]. The DAGAN model exhibits strong generalization across synthetic datasets and real-world conditions, as well as superior visual and quantitative performance.

In Fig 6, DCP [33], DehazeNet [34], and GMAN [35], GCANet [36], and DCPDN [37] produce outputs with darker backgrounds, while fail to completely remove the haze. EPDN [38], HIDEGAN [39], GPGDN [40], and DPTE-Net [41] show color variations and overexposure. The DAGAN model, on the other hand, successfully restores dehazed images while maintaining fine details.

Fig.3. Visualization of RESIDE dataset images results after applying various dehazing algorithms: (a) Haze image, (b) DCP [33], (c) DehazeNet [34], (d) GMAN [35], (e) GCANet [36], (f) DCPDN [37], (g) EPDN [38], (h) HIDEGAN [39], (i) GPGDN [40], (j) DPTE-Net [41], (k) Proposed, and (l) Ground Truth.

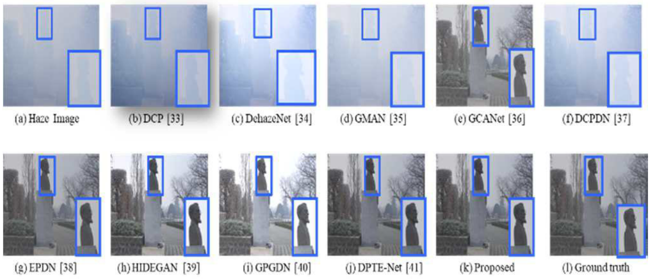

Fig.4. Visualization of Dense-Haze dataset images results after applying various dehazing algorithms: (a) Haze image, (b) DCP [33], (c) DehazeNet [34], (d) GMAN [35], (e) GCANet [36], (f) DCPDN [37], (g) EPDN [38], (h) HIDEGAN [39], (i) GPGDN [40], (j) DPTE-Net [41], (k) Proposed, and (l) Ground Truth, Zoom in to see better visualization.

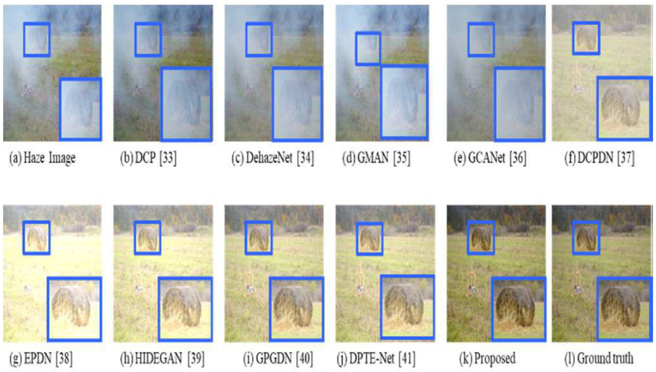

Fig.5. Visualization of NH-Haze dataset images results after applying various dehazing algorithms: (a) Haze image, (b) DCP [33], (c) DehazeNet [34], (d) GMAN [35], (e) GCANet [36], (f) DCPDN [37], (g) EPDN [38], (h) HIDEGAN [39], (i) GPGDN [40], (j) DPTE-Net [41], (k) Proposed, and (l) Ground Truth, Zoom in to see better visualization.

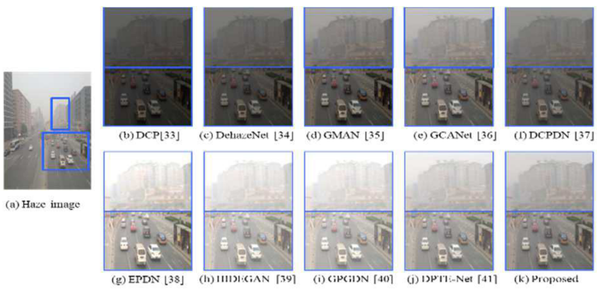

Fig.6. DAGAN and other SOTA techniques are used to visualize the comparison of dehazed and real-haze images. (a) Haze image, (b) DCP [33], (c) DehazeNet [34], (d) GMAN [35], (e) GCANet [36], (f) DCPDN [37], (g) EPDN [38], (h) HIDEGAN [39], (i) GPGDN [40], (j) DPTE-Net [41], and (k) Proposed, Zoom in to see better visualization.

VI. OBJECT DETECTION WITH YOLOV8

The performance of the suggested DAGAN in improving object detection in murky environments is assessed in this section. In tasks like tracking and navigation, haze lowers detection accuracy because most detectors are trained on clear images. After processing DAGAN dehazed outputs using RESIDE test images, YOLOv8 [19] demonstrated improved object localization and increased detection confidence. As Fig. 7 illustrates, DAGAN performs better than other SOTA techniques, proving its resilience in practical situations.

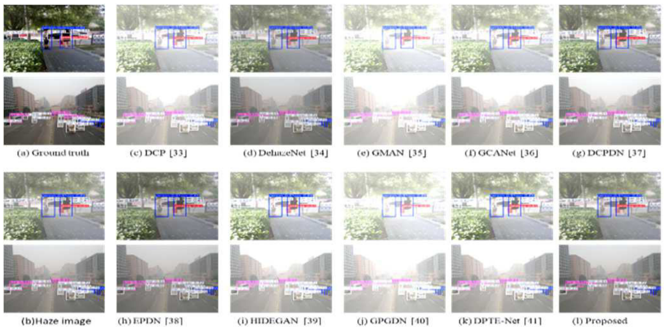

Fig.7. Visualization of two RESIDE images object detection results after applying various dehazing algorithms. (a) Ground Truth, (b) Haze image, (c) DCP [33], (d) DehazeNet [34], (e) GMAN [35], (f) GCANet [36], (g) DCPDN [37], (h) EPDN [38], (i) HIDEGAN [39], (j) GPGDN [40], (k) DPTE-Net [41], and (l) Proposed.

VII. ABLATION STUDY

All of the DAGAN components were tested using ablation experiments. The PSNR dropped by more than 2 dB when the CSAR block was removed, indicating how important the CSAR block is for restoring fine details. In comparison to the standard U-Net, the modified U-Net architecture (with less feature overlap and Leaky ReLU activation) offered a PSNR improvement of about 1.4 dB. Also, the best SSIM score (>0.96) was obtained when adversarial loss and content loss were combined, whereas using either loss alone resulted in noticeably worse SSIM performance. This demonstrates how the design improves and increases the model's ability to remove haze.

VIII. CONCLUSION

This paper presents a novel GAN-based method for image dehazing, combining a patch-level discriminator with a generator inspired by U-Net. The model enhances spatial and contextual information, allowing for the clear restoration of hazy visuals. It shows strong results in PSNR and SSIM evaluations and effectively retains intricate details, including human features. The data highlights the DAGAN model's accuracy in real-world dehazing scenarios and it is potential to support further developments in image restoration.

The system achieves higher object detection accuracy in low visibility situations when paired with YOLOv8. Additionally, the lightweight design enables real-time performance, effectively detecting previously obscured vehicles and pedestrians due to haze. Additionally, even on low-resource hardware.

This technique performs effectively on images affected by glare or fog, making it well-suited for real-time scenarios such as autonomous driving applications where quick and reliable dehazing is essential for safety and performance.

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